

Coke Zero vs Pepsi Zero/Max — Reddit Difference-in-Differences

```
if (!requireNamespace("here", quietly = TRUE))
  install.packages("here", repos = "https://cloud.r-project.org")

library(here)
library(readr)
library(dplyr)
library(stringr)

reddit <- read_csv(here("data", "coke_pepsi_reddit_all.csv"))
glimpse(reddit)
```

Rows: 57,032

Columns: 12

```
$ id      <chr> "5lclxy", "5lcq9w", "5lepql", "5lg02t", "5lk7gp", "5llb40"~
$ type    <chr> "post", "post", "post", "post", "post", "post", "post", "p~
$ subreddit <chr> "proED", "opiates", "proED", "ketogains", "Showerthoughts"~
$ author  <chr> "theirishone", "hiAnxiety", "Castoroilexperiment", "apache~
$ title   <chr> "Starting 2017 with damage control. Any of you all have re~
$ text    <chr> "Today I bought two bottles of caffeine mio and a box of v~
$ score   <dbl> 32, 1, 25, 12, 1, 51, 4, 7, 109, 10, 3, 1, 5, 1, 8, 0, 8, ~
$ comments <dbl> 30, 7, 24, 35, 0, 16, 25, 23, 17, 14, 3, 1, 13, 9, 20, 51,~
$ created_utc <dbl> 1483230709, 1483232371, 1483268044, 1483290925, 1483344559~
$ term     <chr> "coke zero", "coke zero", "coke zero", "coke zero", "coke ~
$ group    <chr> "treatment", "treatment", "treatment", "treatment", "treat~
$ window   <chr> "2017_rebrand", "2017_rebrand", "2017_rebrand", "2017_rebr~
```

Exercise 2(c)

```
reddit |>
  filter(subreddit == "cocacola") |>
  glimpse()
```

Rows: 924

Columns: 12

```
$ id      <chr> "5ppxb6", "6iru5x", "6ppuk2", "6prrw3", "6sp0vb", "6tg86g"~
$ type    <chr> "post", "post", "post", "post", "post", "post", "post", "p~
$ subreddit <chr> "cocacola", "cocacola", "cocacola", "cocacola", "cocacola"~
$ author  <chr> "[deleted]", "Aus_pol", "redbullchugger", "darthmal321", "~
$ title   <chr> "Inside Coke Zero!", "Fresh Coke No Sugar (Coke Zero repla~
$ text    <chr> "[deleted]", NA, NA, "DON'T DISCONTINUE COKE ZERO!! My fam~
$ score   <dbl> 1, 1, 1, 13, 4, 8, 7, 6, 19, 1, 1, 4, 1, 1, 2, 2, 2, 3, 3,~
$ comments <dbl> 0, 0, 0, 6, 1, 3, 12, 6, 9, 0, 0, 7, 0, 0, 5, 6, 5, 6, 2, ~
$ created_utc <dbl> 1485190185, 1498114166, 1501091995, 1501109168, 1502317413~
$ term     <chr> "coke zero", "coke zero", "coke zero", "coke zero", "coke ~
$ group    <chr> "treatment", "treatment", "treatment", "treatment", "treat~
$ window   <chr> "2017_rebrand", "2017_rebrand", "2017_rebrand", "2017_rebr~
```

Exercise 2(d)

```
# (1) Total row count
nrow(reddit)
```

```
[1] 57032
```

```
# (2) Share of posts vs. comments
reddit |>
  count(type) |>
  mutate(share = n / sum(n))
```

```
# A tibble: 2 x 3
  type      n share
<chr> <int> <dbl>
1 comment 51156 0.897
2 post    5876 0.103
```

```
# (3) Date range of created_utc
range(as.POSIXct(reddit$created_utc, origin = "1970-01-01", tz = "UTC"))
```

```
[1] "2017-01-01 00:31:49 UTC" "2021-12-31 23:58:57 UTC"
```

Exercise 2(e)

```
cocacola <- reddit |>
  filter(subreddit == "cocacola") |>
  mutate(date = format(as.POSIXct(created_utc, origin = "1970-01-01", tz = "UTC"), "%Y-%m-%d"))

top3_comments <- cocacola |> filter(type == "comment") |> arrange(created_utc) |> slice(1:3)
top3_posts <- cocacola |> filter(type == "post") |> arrange(created_utc) |> slice(1:3)

bind_rows(top3_comments, top3_posts) |>
  select(type, subreddit, author, title, text, score, date)
```

```
# A tibble: 6 x 7
  type      subreddit author          title          text    score date
<chr>    <chr>      <chr>        <chr>        <chr> <dbl> <chr>
1 comment cocacola  XenobiaXD    <NA>          "I k~      3 2017~
2 comment cocacola  JustGoingWithIt <NA>          "\"R~      1 2017~
3 comment cocacola  PM_ME_YOUR_MAUSE <NA>          "I s~      2 2017~
4 post    cocacola  [deleted]    Inside Coke Zero! "[de~      1 2017~
5 post    cocacola  Aus_pol      Fresh Coke No Sugar (Cok~ <NA>      1 2017~
6 post    cocacola  redbullchugger Coca-Cola is killing Cok~ <NA>      1 2017~
```

```
print_rows <- function(df, section_title) {
  cat(sprintf("### %s\n\n", section_title))
  for (i in seq_len(nrow(df))) {
    r <- df[i, ]
    raw_text <- if (is.na(r$text)) "" else r$text
    body <- if (nchar(trimws(raw_text)) == 0) "[no text]" else raw_text
    score <- if (is.na(r$score)) "NA" else as.character(r$score)
    title <- if (is.na(r$title)) "" else r$title
    if (r$type == "post" && nchar(trimws(title)) > 0) {
      cat(sprintf("**%s | u/%s | r/%s | score: %s | %s**\n\n",
                  title, r$author, r$subreddit, score, r$date))
    } else {
```

```

        cat(sprintf("**[comment] u/%s | r/%s | score: %s | %s**\n\n",
                    r$author, r$subreddit, score, r$date))
    }
    cat(body, "\n\n---\n\n")
}
}

print_rows(top3_comments, "Earliest 3 Comments - r/cocacola")

```

Earliest 3 Comments — r/cocacola

[comment] u/XenobiaXD | r/cocacola | score: 3 | 2017-03-07

I know I would. Drinking some Coke Zero out of my coke cooler while sitting in my coke camping chair, fishing with my coke fishing rod (if that exists I'll buy it too) catching fish with my coke lure and worms dipped in coke.

[comment] u/JustGoingWithIt | r/cocacola | score: 1 | 2017-04-30

“Real” Coke should be made with sugar, not high fructose corn syrup. Coke Zero tastes better, no corn syrup to sweeten it. They use aspartame. However, Mexico’s Coke in the glass bottle is made with sugar-and that has a red label.

Edit: The original contained cocaine, which is the ‘Coca-’ in the name, but when it was outlawed the recipe changed. John Pemberton-creator of Coca-Cola, marketed it first a Pemberton’s French Wine Cola. No one alive today knows what that tastes like. It was much different and more pure.

[comment] u/PM_ME_YOUR_MAUSE | r/cocacola | score: 2 | 2017-07-26

I second this. Nobody touches my damn Cherry Coke Zero

```

print_rows(top3_posts, "Earliest 3 Posts - r/cocacola")

```

Earliest 3 Posts — r/cocacola

Inside Coke Zero! | u/[deleted] | r/cocacola | score: 1 | 2017-01-23

[deleted]

Fresh Coke No Sugar (Coke Zero replacement) Review from Australia | u/Aus_pol
| r/cocacola | score: 1 | 2017-06-22

[no text]

Coca-Cola is killing Coke Zero and replacing it with a new drink | u/redbullchugger
| r/cocacola | score: 1 | 2017-07-26

[no text]

Exercise 3(a)

```
# Save original text before cleaning
text_orig <- reddit$text

clean_text <- function(x) {
  x |>
  # Markdown links [text](url) - keep inner text, drop URL
  str_remove_all("\\\\([\\^\\]]+\\\\)\\\\([\\^]+\\\\)") |>
  str_replace_all("\\\\([\\^\\]]+\\\\)\\\\([\\^]+\\\\)", "\\1") |>
  # Bare URLs
  str_remove_all("https?:\\/\\/\\\\S+") |>
  # User and subreddit mentions
  str_remove_all("(?i)\\\\bu\\/\\\\S+") |>
  str_remove_all("(?i)\\\\br\\/\\\\S+") |>
  # Emphasis chars (**, *, _, `) - keep inner text
  str_remove_all("[*_`]+") |>
  # Header chars at line start
  str_remove_all("(?m)^(#+\\\\s*)" |>
```

```

# HTML entities
str_replace_all("&", "&") |>
str_replace_all("<", "<") |>
str_replace_all(">", ">") |>
str_replace_all(""", "\"") |>
str_replace_all(" ", " ") |>
# Collapse runs of whitespace (preserving line breaks is fine)
str_replace_all("[ \\t]+", " ") |>
str_trim()
}

chars_before <- sum(nchar(reddit$text), na.rm = TRUE) +
  sum(nchar(reddit$title), na.rm = TRUE)

reddit <- reddit |>
  mutate(
    text = clean_text(text),
    title = clean_text(title)
  )

chars_after <- sum(nchar(reddit$text), na.rm = TRUE) +
  sum(nchar(reddit$title), na.rm = TRUE)

cat(sprintf(
  "Characters before cleaning: %s\nCharacters after cleaning:  %s\nCharacters removed:
  format(chars_before, big.mark = ","),
  format(chars_after,  big.mark = ","),
  format(chars_before - chars_after, big.mark = ",")
))

```

```

Characters before cleaning: 39,224,391
Characters after cleaning:  35,434,133
Characters removed:        3,790,258

```

Exercise 3(b)

```

# Pass 1: cross-posting - same text + author across any subreddit
n_before <- nrow(reddit)
reddit   <- distinct(reddit, text, author, .keep_all = TRUE)
n_after  <- nrow(reddit)

```

```
cat(sprintf("Pass 1 (cross-posting: same text + author)\n Before:  %d\n After:  %d\n Dropped: %d\n",  
            n_before, n_after, n_before - n_after))
```

Pass 1 (cross-posting: same text + author)

Before: 57032

After: 55693

Dropped: 1339

```
# Pass 2: coordinated inauthentic behavior - same text + subreddit across any author  
n_before <- nrow(reddit)  
reddit    <- distinct(reddit, text, subreddit, .keep_all = TRUE)  
n_after   <- nrow(reddit)  
cat(sprintf("Pass 2 (coordinated: same text + subreddit)\n Before:  %d\n After:  %d\n Dropped: %d\n",  
            n_before, n_after, n_before - n_after))
```

Pass 2 (coordinated: same text + subreddit)

Before: 55693

After: 54445

Dropped: 1248

```
# Write cleaned corpus for Part 3  
write_csv(reddit, here("data", "coke_pepsi_reddit_cleaned.csv"))  
cat(sprintf("Wrote %d rows to data/coke_pepsi_reddit_cleaned.csv\n", nrow(reddit)))
```

Wrote 54445 rows to data/coke_pepsi_reddit_cleaned.csv